Research Practice Report

Mechanisms and Robotics

BITS G540

"Design and Early Stage Development of a Tendon-Driven Underactuated Robotic Hand with Future EMG-Based Control "

Submitted

by

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7th/May/2026

PROJECT REPORT

RESEARCH PRACTICE

Problem Statement · Vision · Mission · Research Direction

Design and Early-Stage Development of a Tendon Driven Underactuated Robotic Hand with Future EMG Based Control

1. Problem Statement

Upper limb loss significantly affects a person's ability to perform daily tasks. While commercial myoelectric prosthetics exist, they remain expensive, mechanically fragile under real-world use, and often too complex for patients in resource-constrained environments to maintain or afford. Research-grade prosthetic hands with high DOF tend to be overengineered for clinical practicality, while low-cost designs frequently compromise on grasp diversity or mechanical reliability.

The core engineering challenge is not simply building a robotic hand, it is building the right kind of hand: one that is mechanically robust, sufficiently dexterous, affordable to produce and repair, and realistic to integrate with physiological control signals like EMG.

This project addresses three interconnected problems:

- **Mechanical Design Gap:** There is no clear consensus on the simplest tendon-driven, underactuated architecture that achieves practical grasp coverage without excessive actuator count. Existing open-source designs either prioritize dexterity at the cost of complexity (ORCA V1) or simplicity at the cost of grasp range (basic gripper systems). A middle ground modular, low-actuator, adequate-DOF needs to be studied, selected, and developed systematically.
- **Prototype Development:** Most robotic hand designs available in literature are either fully realised research platforms (requiring institutional resources) or toy-level builds (lacking mechanical reliability). There is a gap in honest, documented iterative development: studying which design problems arise at the prototype stage and how they are resolved.
- **Control Integration Readiness:** EMG-based control is frequently discussed in prosthetic hand literature, but the control system is often developed in isolation from the mechanical system. In practice, an unstable or unreliable mechanical platform cannot benefit from a sophisticated EMG controller. The sequence matters mechanical foundations must be established before control integration becomes meaningful.

This work directly addresses the mechanical design and prototype development phases, while laying the groundwork for future EMG and intelligent control integration.

2. Research Vision

The long-term vision of this project is to develop a low-cost, modular, underactuated tendon-driven prosthetic robotic hand that is mechanically reliable, practically dexterous, and capable of intelligent grasp selection through EMG-based control and machine learning, designed to be manufacturable, maintainable, and adaptable in resource-limited settings.

This vision consists of a layered philosophy: mechanical intelligence first, electronic intelligence second.

The hand should rely primarily on its physical design, tendon routing, joint compliance, and underactuated synergy to perform adaptive grasping without complex real-time computation. Once this mechanical foundation is robust and repeatable, a lightweight EMG-based control layer can be added that selects and triggers grasps based on the user's residual muscle signals. Finally, a machine learning layer can refine grasp selection over time using signal pattern recognition.

The goal is not a hand that does everything. The goal is a hand that does the most common things reliably, can be built and repaired affordably, and grows in capability as the control layer matures.

3. Mission Statement

To systematically design, prototype, and develop a tendon-driven underactuated robotic hand platform through iterative engineering, starting with mechanical robustness and grasp functionality, progressing to EMG-based servo control, and ultimately toward intelligent adaptive grasp behaviour, with each phase building on validated results from the previous one.

Specifically, the mission involves:

- Identifying and selecting the most suitable open-source robotic hand architecture for iterative low-cost development.
- Building and refining a mechanically reliable tendon-driven hand prototype with a practical 5-DOF configuration.
- Validating grasp coverage against standard human grasp taxonomies.
- Integrating EMG signal acquisition and developing a servo control interface once mechanical reliability is confirmed.
- Establishing a clear, documented development path from basic actuation to intelligent grasp control.

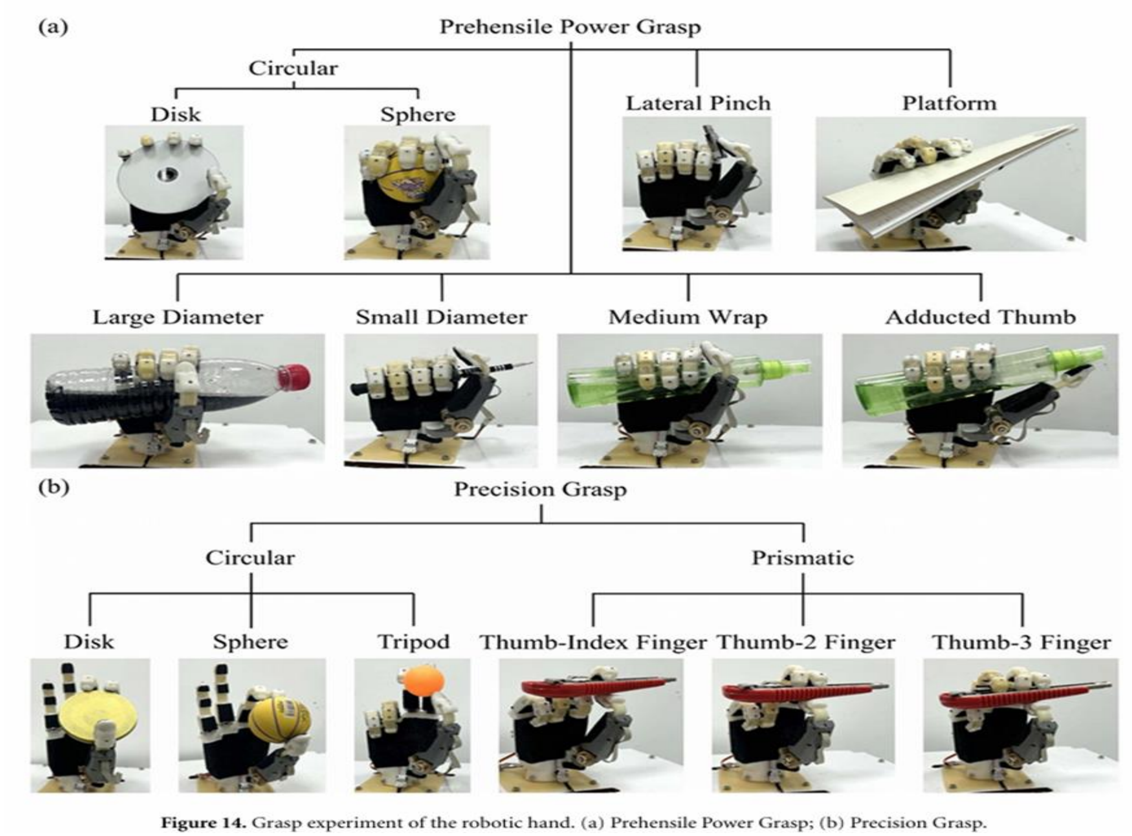

Figure: *Goal to achieve all these 13-grip patterns with 5-DOF*

4. Current Project Status

The project is currently in Phase 1: Mechanical Foundation Development. It is approximately 70% complete. This section documents what has been done, what is in progress, and what has been explicitly deferred.

4.1 Completed Work

Literature and Design Exploration

Four open-source robotic hand architectures were studied in detail: Aero Hand Open (TetherIA), Tact Hand (low-cost prosthetic), ORCA V1 (ETH Zurich), and the Mahdi Bionic Hand. CAD models were imported and analyzed for joint structure, tendon routing geometry, and prototyping feasibility. Each design was evaluated against criteria including actuation type, under actuation support, modularity, cost, and EMG compatibility.

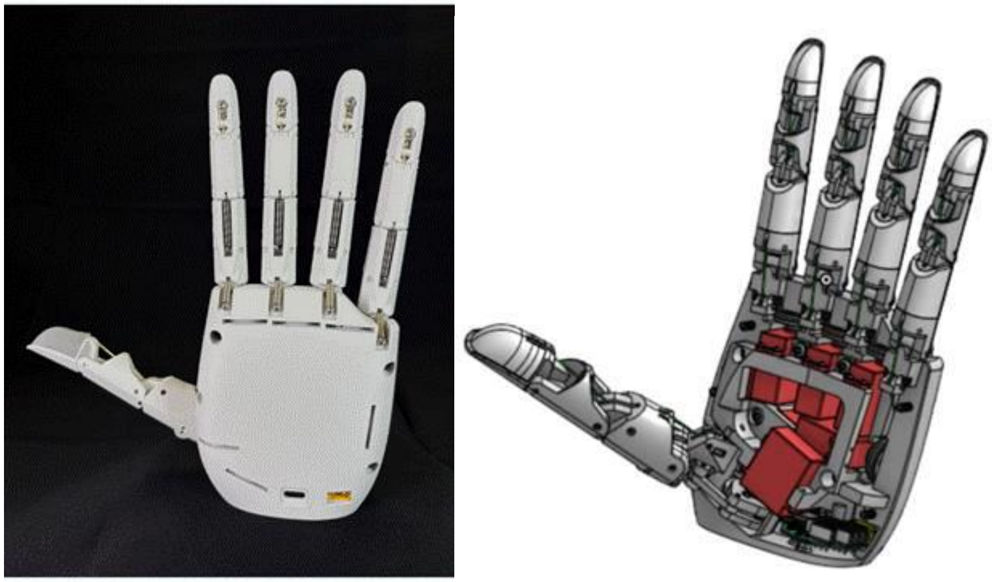

Figure: *Aero Hand Open - Assembled*

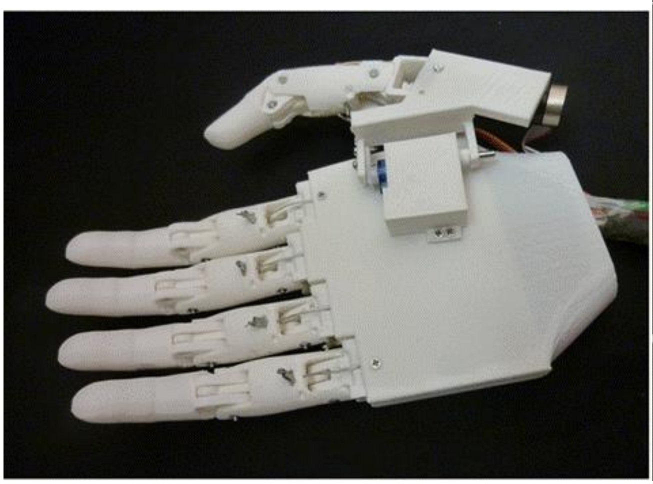

Figure: *Tact: Low-cost, Advanced Prosthetic Hand*

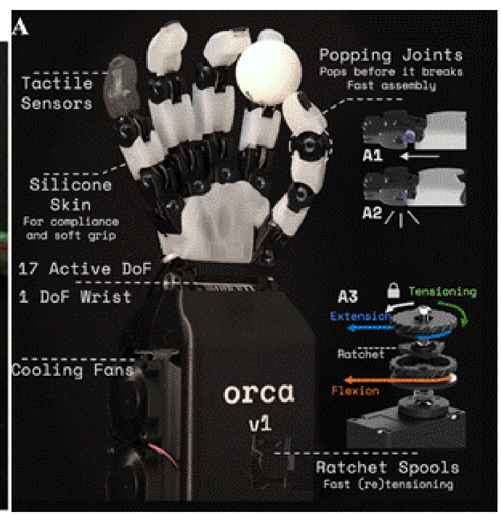

Figure: *The Orca V1 Hand*

Design Selection

The Mahdi Bionic Hand was selected as the development platform. The selection was based on its fully 3D-printable modular architecture, detachable finger modules, single-actuator tendon-driven simplicity per finger, under-actuation readiness, and explicit myoelectric compatibility in its original design intent.

Figure**:** *Myoelectric Prosthetic Arm (Mahdi design) ; Final selected one*

Figure*: CAD model of selected Prosthetic Arm design*

Mechanism and Grasp Study

Tendon-driven finger mechanisms were studied across three categories: constrained (one motor per joint), under-constrained (one tendon across multiple joints), and soft/deformable. The under-constrained approach was selected for the four fingers due to its simplicity and adaptive grasp behavior. The thumb mechanism was studied separately; a 2-DOF worm-gear-based transmission providing independent abduction and flexion control was identified from literature as the appropriate approach.

Grasp analysis was performed using Napier's classification and the Feix GRASP taxonomy. A proposed 5-DOF architecture, 2-DOF thumb, independent index and middle fingers, and combined ring-little actuation via adaptive synergy was shown to support 13 of the key grasp types in daily use.

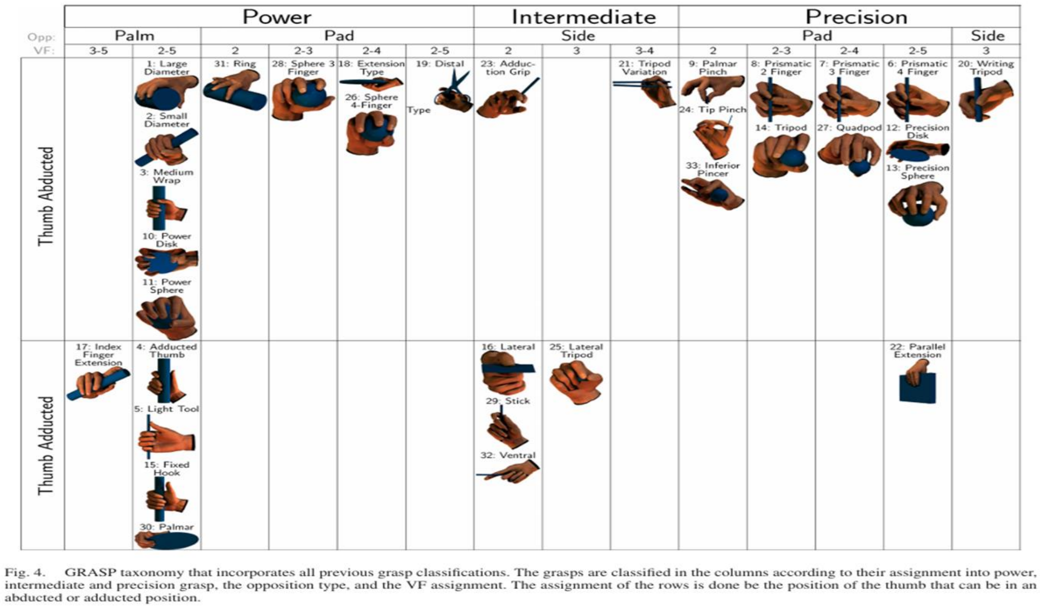

Figure: *Grasp taxonomy used for target grip selection*

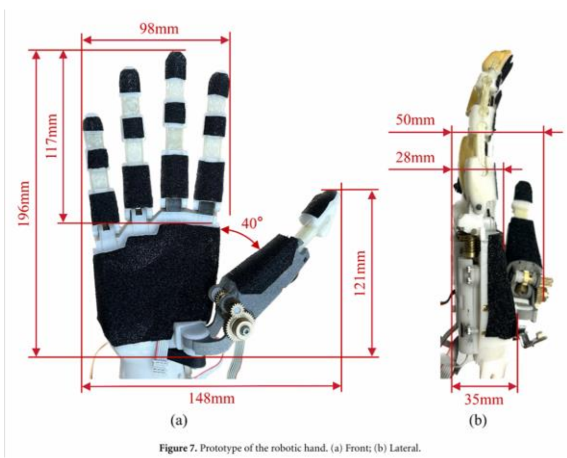

Figure: *2-DOF thumb positioning*

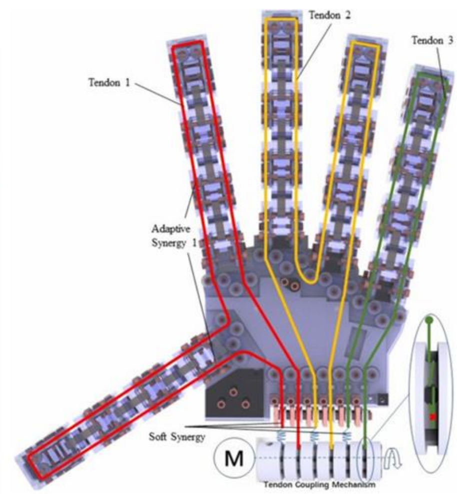

Figure: *Ring + Little finger mechanism*

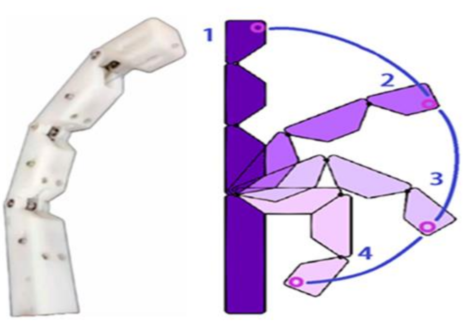

Figure: *Under-actuated finger mechanism*

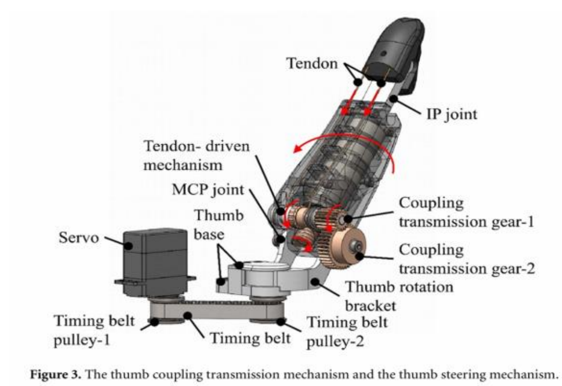

Figure: *Thumb mechanism*

Criterion	Aero Hand Open	Tact Hand	ORCA V1	Mahdi Bionic Hand
Actuation	6 independent motors	Servo-driven tendons	3 motors/finger	Single servo per finger
Mechanism	Tendon-driven + spring	Tendon + linkage hybrid	Tendon + rigid links	Tendon-driven modular
Under actuation	No	Partial	No	Yes
3D Printable	Partial	Yes	Partial	Fully printable
Modularity	Low	Medium	Low	High
Prototyping Ease	Moderate	Moderate	Complex	High
EMG Compatibility	Possible	Possible	Complex	Designed for it
Cost	Medium-High	Low-Medium	High	Low
Selected	No	No	No	Yes

***Table**: Comparative Analysis of Existing Designs*

Early Prototype Development

3D printing of finger modules was initiated. Servo motors were selected and integrated with a microcontroller for tendon actuation. Badminton racket string was used as the tendon material for initial testing due to its low stretch and appropriate tensile properties. An SMPS power supply was procured and integrated into the power circuit. Basic servo control logic was implemented and tested. Tendon-driven finger flexion and extension were observed in initial testing runs.

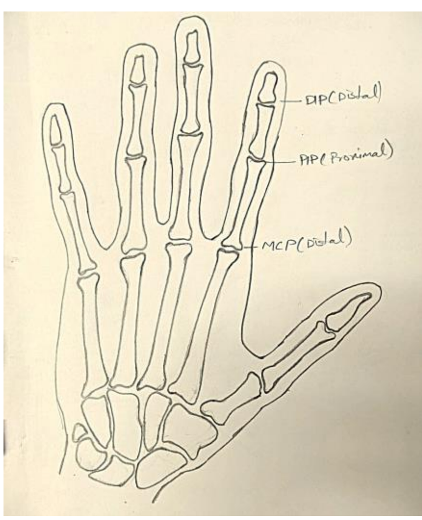

***Figure:** Normal human left-hand* ***Figure:** Final fabricated Hand*

Hardware and Material Selection

The current prototype uses a deliberately simple hardware stack so that mechanical problems can be identified without hiding them behind software compensation. The main materials and components are PLA printed structural parts, badminton thread/string for tendon trials, brass fasteners for the finger and linkage assemblies, a microcontroller for basic servo commands, and a 12V 5A SMPS for actuator power during bench testing.

PLA was selected for the first mechanical iterations because it supports fast 3D printing, low-cost redesign, and easy replacement of failed parts. Its limitations are also clear: small joint features can crack, holes can wear with repeated assembly, and tolerances vary with print orientation and printer settings. These limits are being treated as part of the mechanical refinement process rather than ignored.

The badminton thread/string is not considered the final tendon material. It is being used as a practical early test material for routing, friction, knotting, and tensioning studies. A more controlled tendon material may be selected later, after the routing geometry and return mechanism are stable.

Servo Motor Selection

The Waveshare ST3235 serial bus servo was selected for tendon actuation because it provides high torque output (20 kg·cm), a compact form factor, and closed-loop position feedback. The servo uses a coreless DC motor, which supports faster acceleration and smoother response than a conventional brushed hobby servo. Its integrated magnetic encoder and serial communication interface provide position and load feedback, making it suitable for future closed-loop prosthetic control experiments. For the current phase, its torque capability is considered sufficient for tendon-driven finger flexion and basic grasp stabilization during prototype testing.

4.2 Ongoing Work

The current development focus is on mechanical refinement rather than functional expansion. Specific areas being worked on:

- **Mechanical robustness:** Some printed joints showed minor cracking or misalignment under repeated actuation. These are being redesigned.
- **Tendon routing:** The tendon path through joint channels needs refinement to reduce friction and prevent cable derailing under high tension.
- **Assembly reliability**: Fastener selection and joint tolerances are being adjusted to improve assembly repeatability.
- **Grasp behavior improvement:** Initial tests showed inconsistent finger closure. Tendon attachment points and elastic return tension are being adjusted.

4.3 Engineering Observations

The main learning from the current prototype is that the mechanical layer controls the quality of every later control result. Servo commands can move the tendon, but the final finger motion depends strongly on tendon path, joint friction, return elasticity, and assembly tolerance.

- **Tendon friction:** Small bends and rough printed channels increase resistance and reduce repeatability. This is most visible when the finger begins to close from the fully open position.
- **Tendon derailing:** The tendon can leave the intended path when tension is high or when the routing channel does not guide it through the joint axis cleanly.
- **Joint cracking and wear:** Some PLA joint regions and thin features show minor cracking after repeated assembly or actuation. These areas need thicker sections, better print orientation, or revised fastener placement.
- **Finger closure inconsistency:** Identical servo angles do not always produce identical fingertip positions. The cause appears to be a combination of tendon stretch, friction, joint play, and elastic return balance.
- **Tolerance sensitivity:** small changes in hole diameter, pin fit, and screw tightening affect joint motion. Repeatable assembly is therefore as important as the CAD geometry.
- **Servo placement influence:** Servo position changes tendon entry angle, available pull length, and friction. The actuator layout must be treated as part of the mechanism design, not only an electronics placement problem.
- **Elastic return balancing:** Return force must be high enough to open the finger, but not so high that the servo wastes torque overcoming the return element. This balance is still being tuned.
- **Underactuated motion tradeoff:** Under actuation reduces actuator count and supports passive adaptation, but it also reduces independent joint control. The design must accept this tradeoff instead of trying to behave like a fully actuated robotic hand

4.4 Explicitly Deferred Work

The following items are intentionally not being worked on in the current phase. They require the mechanical platform to be stable before they add any value:

NOT STARTED	Full prosthetic hand assembly: the hand is not yet fully assembled as an integrated system.

NOT STARTED	EMG-based control EMG hardware (DELSYS sensor) is available, but signal integration with servo control has not begun. This is deferred until mechanical actuation is reliable.

NOT STARTED	ANN/ML-based grasp classification. No machine learning has been implemented or trained.

NOT STARTED	Grip library, tactile feedback, and autonomous grasp adaptation are all long-term goals with no current implementation.

5. Research Direction and Development Roadmap

The development is structured in four sequential phases. Each phase has a defined completion criterion before the next phase begins. This prevents the common failure mode of attempting advanced control integration before the mechanical platform is ready for it.

Phase	Focus Area	Key Activities	Status
Phase 1 (Current)	**Mechanical Foundation**	Design selection · 3D printing finger modules · Servo + tendon integration · Basic actuation testing · Mechanical debugging	**70% Done**
Phase 2 (Near-term)	**Stable Hand Platform**	Full 5-DOF assembly · Tendon routing refinement · Grasp stability testing · Servo repeatability validation · Component integration	**Planned**
Phase 3 (Mid-term)	**EMG Integration**	DELSYS EMG setup · Signal preprocessing · Threshold-based open/close control · Linear regression mapping · Multi-channel EMG control	**Planned**
Phase 4 (Long-term)	**Intelligent Control**	Grip library development · ANN-based grasp classification · Adaptive grasp behavior · User-specific calibration · Force/tactile feedback	**Future**

5.1 Phase 1 Mechanical Foundation (Current, 70% Complete)

The immediate priority is a mechanically stable and repeatable hand platform. This means:

- All five finger DOFs actuate correctly and repeatedly without mechanical failure.
- Tendon routing is clean, no cable friction spikes, no derailing under load.
- The thumb achieves full abduction and flexion range with the worm-gear mechanism.

- The ring and little finger combined mechanism closes smoothly and adapts to object contact.
- At least five grasp configurations can be demonstrated reliably with physical objects.

Completion benchmark: The hand can perform five target grasps (large power wrap, lateral pinch, tripod, small diameter, medium wrap) consistently across 10 successive trials without mechanical failure.

5.2 Phase 2 Stable Platform and Grasp Validation

Once the finger mechanisms are robust, the full hand will be assembled with all five DOFs integrated. Grasp coverage will be experimentally validated; each of the 13 target grasp types will be tested with representative objects. Servo repeatability will be quantified. This phase produces the first fully functional prototype.

- Full 5-DOF assembly on a palm structure.
- Grasp coverage test: 13 grasp types validated with physical objects.
- Servo position repeatability measured across 20 cycles per grasp.
- Structural load test: basic grip force measurement.

Completion criterion: All 13 target grasps achieved with physical objects. Servo repeatability within ±5° across 20 trials.

5.3 Phase 3 EMG Integration

The DELSYS EMG sensor will be used to acquire signals from the forearm. Initial integration will be simple and direct:

- **Threshold control:** above a signal threshold, the hand closes; below it, the hand opens. This validates the EMG-to-servo interface.
- **Proportional control:** EMG amplitude is mapped to servo angle through a linear regression model ($Y = mX + C$). This allows proportional finger closure.
- **Multi-channel mapping:** four EMG channels mapped to four servo channels, enabling partial independent finger control from muscle signals.

The key principle of this phase is to keep the EMG processing simple and the feedback loop short. Advanced signal processing and classification are deferred to Phase 4.

Completion criterion: The user can open and close the hand using forearm muscle contraction. Proportional control achieves at least three distinct grip levels (open, half, closed) reliably.

5.4 Phase 4 Intelligent Grasp Control (Long-Term)

This phase introduces machine learning into the control pipeline. The goal is to grasp intent recognition. The system identifies which grasp the user intends from the pattern of their EMG signals and triggers the corresponding pre-programmed servo configuration.

- **EMG data collection:** labelled datasets for each target grasp type collected from users.
- **ANN-based classifier:** trained to map EMG feature vectors to grasp class labels.
- **Grip library:** pre-stored servo angle configurations for each grasp type, triggered by classifier output.
- **Size adaptation:** grip closure depth adjusted based on EMG signal amplitude
- **User-specific calibration**: model fine-tuned to an individual user's EMG patterns.
- Semi-adaptive behaviour: mechanical synergy handles final finger conformation; ANN handles grasp selection.

This phase represents the convergence of mechanical intelligence (adaptive tendon compliance) and artificial intelligence (learned grasp intent recognition). It is a long-term goal and will require validated results from all preceding phases.

Completion benchmark: Demonstration of EMG-based grasp selection using a predefined grip library and basic pattern classification methods.

6. Research Positioning

This project is positioned as an iterative research and engineering effort, not a finished product, not a complete system, and not a demonstration of novel algorithmic invention. The contribution at each phase is:

Phase 1	A documented, honest comparative design study and early prototype development process for a low-cost tendon-driven robotic hand with clear mechanical trade-off analysis.

Phase 2	A validated, physically functional 5-DOF underactuated prosthetic hand prototype with documented grasp coverage and mechanical reliability data.

Phase 3	A functional EMG-to-servo control interface, validated on a stable mechanical platform, demonstrating proportional and multi-channel muscle control.

Phase 4	An integrated intelligent prosthetic hand system combining adaptive tendon mechanics with ANN-based grasp intent recognition, calibrated per user.

Each phase is independently valuable and independently publishable. The project does not require completion of Phase 4 to have made a research contribution.

"The project currently focuses on the mechanical development and refinement of a tendon-driven robotic hand platform before moving toward EMG-based control and intelligent grasp adaptation.

The project follows an iterative engineering approach where mechanical reliability is prioritized before advanced control integration. "

7. References

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